BRIDGES IN MATHEMATICS • GRADE 2 • UNIT 3

Addition & Subtraction Within 100: Jump & Split Strategies

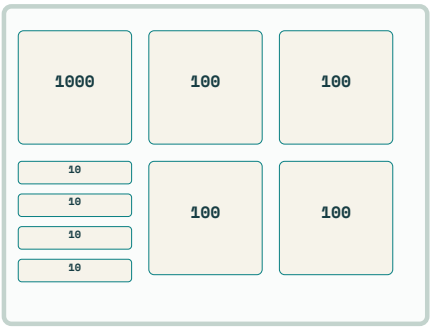
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.2.NBT.B.5

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Place Value Base Ten

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Split 35 into 3 tens rods and 5 unit cubes in the Place Value Tray. Add tens first, then ones.

Sketch Model on Paper:
Draw an open number line: start at 48, jump +30 to 78, jump +5 to 83.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	48	X: 48	Start at 48
2.	ENTER	Y: 48 | X: 48	Stack lift
3.	30	Y: 48 | X: 30	Key tens jump (+30)
4.	+	X: 78	After tens jump: 78

EXPECTED LCD X: 83 (48 + 30 + 5)

1 **Solve 45 + 27 using jump strategy: 45 + 20 + 7.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Solve 38 + 34 using jump strategy: 38 + 30 + 4.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Solve 58 - 24 using jumps backwards: 58 - 20 - 4.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Solve 83 - 36 using jumps backwards: 83 - 30 - 6.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 3

Addition & Subtraction Within 100: Jump & Split Strategies • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Place Value Base-Ten Studio (BASE-10)

CAD Blueprint: place_value_base_ten

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/place_value_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Place Value Tray & Base-Ten Blocks (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 45 + 27 using jump strategy: 45 + 20 +** Standard: CCSS.MATH.2.NBT.B.5	72	[4][5] [ENTER] [2][0] [+] [7] [+]
2	**Solve 38 + 34 using jump strategy: 38 + 30 +** Standard: CCSS.MATH.2.NBT.B.5	72	[3][8] [ENTER] [3][0] [+] [4] [+]
3	**Solve 58 - 24 using jumps backwards: 58 - 20** Standard: CCSS.MATH.2.NBT.B.5	34	[5][8] [ENTER] [2][0] [-] [4] [-]
4	**Solve 83 - 36 using jumps backwards: 83 - 30** Standard: CCSS.MATH.2.NBT.B.5	47	[8][3] [ENTER] [3][0] [-] [6] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure students add the tens jump first before adding the ones jump.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.